Improving Multiple-Choice Question Performance in LLMs via Activation Steering of Query Styles

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Abstract

Recent work "Mind Your Tone" has demonstrated that prompts varying in politeness, urgency, or rudeness, improve Large Language Model (LLM) performance on reasoning tasks. However, these results have been shown through prompting in a "black-box" evaluation setting. Separately, activation steering has emerged as a powerful method for controlling model behaviour, though it has been predominantly used for high-level traits such as safety and refusal. In our work, we bridge these two areas by applying activation steering to low-level stylistic behaviours (politeness, urgency, rudeness, etc.) as an effective inference-time approach to improve Multiple-Choice Question performance. Our results show that steering via learned persona vectors achieve performance gains of up to 12.5% across diverse benchmarks (Mind Your Tone, MMLU Math, TruthfulQA, RACE). Beyond accuracy improvements, we find that our method acts as a focusing mechanism, reducing entropy and Expected Calibration Error (ECE). We provide a detailed analysis of optimal steering configurations and demonstrate that these persona vectors generalize effectively across benchmarks from different domains.

1 Introduction

Chat-based interfaces have become the dominant way of interacting with Large Language Models (LLMs), with users using a diverse set of conversational styles (Bai et al., 2022; Demszky et al., 2020). Queries often carry distinct stylistic attributes, such as being polite, urgent, or insulting. This has motivated a significant body of work on how LLMs adopt or express conversational personas (Truong et al., 2025; Ge et al., 2025; Truong et al., 2025; Li et al., 2025). To control model behaviour, researchers developed techniques, such as prompt engineering and activation steering, to either elicit desired behaviour or restrain unwanted behaviours of LLMs.

We first outline the background work on prompt engineering and activation steering, then highlight recent findings on how how varying levels of prompt politeness affects models' downstream task performance. Finally, we present our work, showing how to improve task performance using activation steering via learned persona vectors.

Our implementation code is publicly available at [1]

1.1 Background and Related Work

1.1.1 Prompt Engineering

One of the ways for modulating LLM behaviour is prompt engineering, which involves crafting natural language instructions to alter the textual input for the model. This includes techniques such as "persona adoption" (White et al., 2023) where the model is told to act as a specific character or professional. While effective for simple alignment, prompt engineering relies on the model's ability to adhere to linguistic instructions. This can lead to inconsistent results, as models may ignore (Qi et al., 2024) instructions or adhere in a limited way.

1.1.2 Activation Steering and Model Alignment

Activation steering is another alternative for guiding model behaviour. Unlike prompt engineering, which only modifies the model's inputs, activation

[1] https://github.com/oszkarurban/l101-activationsteering/

steering operates directly on the model's internal representations. This technique involves identifying directions within the model's hidden activations that correspond to target concepts and adding these vectors to the residual stream at inference time. (Turner et al., 2024; Panickssery et al., 2024; Templeton et al., 2024) This allows for more granular and mechanistic control over the model's behaviour than prompt engineering.

Research in this area (Chen et al., 2025; Arditi et al., 2024), however, has focused primarily on high-level behaviour such as alignment and safety. They used activation steering to mitigate sycophancy, the tendency of models to agree with user misconceptions, and to enforce safety guardrails. While these studies demonstrate the power of activation steering for "high-level" traits, they don't explore how "low-level" conversational behaviour, like politeness or rudeness, influence model's reasoning processes and its performance on downstream tasks.

1.1.3 Stylistic Influence on Reasoning (Mind Your Tone)

The influence of changing the politeness of prompts on task performance was recently explored in Mind Your Tone (Dobariya and Kumar, 2025). This work demonstrated that low-level stylistic behaviours improve GPT-4o model's accuracy on Multiple-Choice Question (MCQ) tasks by up to 4%. These results were achieved by prepending stylistic modifiers, such as polite or insulting prefixes, to queries.

However, Mind Your Tone is limited to a "black-box" evaluation study using the GPT-4o API, focusing only the accuracy changes. Nonetheless, a significant finding of this work is that low-level conversational style is sufficient to improve a model's reasoning and downstream task performance.

1.2 Our Work

Our work bridges these directions by applying activation steering to elicit low-level stylistic behaviours, such as politeness, urgency, and rudeness, that better reflect natural user-LLM interactions. Rather than prompting these styles through textual input, we manipulate the model's latent activations at inference time by steering along learned stylistic directions.

We develop a pipeline to extract persona vectors from internal activations and steer along these vectors to improve performance on multiple-choice tasks. By systematically varying steering layers l and steering strength (α), we identify the optimal layer-(α) configurations for each style and show that effective steering acts as a focusing mechanism, reducing output entropy and Expected Calibration Error.

The main contributions of our work are:

1. We develop an activation-steering pipeline that improves LLM performance on MCQ tasks by steering internal activations along learned vectors representing low-level stylistic behaviours.

2. We show that our solution improves accuracy by up to 12.5% on benchmarks like MMLU Math, TruthfulQA, Mind Your Tone, RACE

3. We show that optimal steering reduces entropy and Expected Calibration Error, thereby refining model confidence and calibration.

2 Methodology

Our approach builds on activation steering by extracting activation directions corresponding to specific persona styles and to manipulate LLM responses by steering along these learned directions at inference time, akin to (Allbert et al., 2025). The methodology consists of two stages.

First, we extract persona vectors; vectors in the model's residual stream that correspond to activations triggered when interacting with the LLM in specific query styles, such as insulting or polite.

In the second stage, we apply activation steering at inference time, injecting these learned vectors to modulate the model's internal states with the goal to improve multiple-choice question performance.

2.1 Defining Persona Styles

We define a set of six distinct persona styles: *Polite*, *Insulting*, *Admiration*, *Urgent*, *Not Urgent*, and *Nervous*. These styles were selected to probe three distinct aspects of interaction:

- **Social aspect (*Polite, Insulting, Admiration*):** These styles aim to alter the general behavioural tone. *Admiration* is designed to

boost the model's ability by providing positive, empowering reinforcement, asserting that the model is highly capable of finding the correct answer. Conversely, *Insulting* acts as the opposite, doubting the model's intellect to see if negative pressure degrades reasoning performance. *Polite* serves as a courteous, respectful tone.

- **Temporal Constraints (*Urgent, Not Urgent*):** These styles manipulate the time aspect of the query. By framing the request as time-critical (*Urgent*) versus low-priority (*Not Urgent*), we investigate whether the model's reasoning capabilities can be modulated by a perceived sense of hurry.
- **Emotional Affect (*Nervous*):** This style is designed as a form of emotional blackmailing. By presenting a highly anxious persona that is desperate for the correct answer, we examine if the model can achieve higher performance through the simulation of high-stakes emotional pressure.

Following best-practice from work on LLM persona-styles (Truong et al., 2025; Ge et al., 2025), we used an LLM (GPT-4.1) to generate the persona-specific prefixes based on descriptions of the aforementioned personalities. To ensure high-quality separation between styles, we later manually reviewed the generated query styles.

Examples of these prefixes include:

- **Admiration:** "You are incredibly smart, and I know you can answer this question:"
- **Insulting:** "Try not to fail again, you *** loser, and answer this question:"
- **Polite:** "May I politely ask you to answer this question:"

Each stylized query is formed by prepending the persona-specific prefix to the original question. This ensures the model's internal states are influenced by the persona style when processing the question. For example:

> **Original Question:** What is the capital of France?
> **With Persona Prefix (Admiration):** You are incredibly smart, and I know you can answer this question: What is the capital of France?

2.2 Extracting Persona Vectors

To represent specific style S, we employ an activation extraction method. We separate our dataset into a Steering split for vector extraction and a held-out Evaluation split for the downstream performance testing.

For each query in the Steering split, we perform a generation pass using both the persona-specific query and the neutral query. To ensure the resulting vector captures the underlying persona style rather than specific phrasing of a single prompt, we generate K ($K = 3$) distinct persona-specific prefixes for every style. We capture the hidden state activations at layer l for each position in the generated sequence, then compute the mean vector across these positions and across the K prefix variations, to produce a robust representation for each style. This design choice was validated through preliminary experiments, which indicated that activations extracted from response phase (corresponding to the "manifestation" of the persona) yielded significantly higher downstream steerability than those extracted including also the prompt tokens (the instruction processing phase). This approach aligns with prior work (Chen et al., 2025) suggesting that the model's behavioural realization is best captured during the generation phase.

Formally, let $H_{S,l} \in \mathbb{R}^{n \times d}$ be the set of collected response activation vectors for style S at layer l, where n is the number of samples and d is the hidden dimension. We compute the mean activation for the styled, $\mu_{S,l}$, and neutral conditions, $\mu_{Neutral,l}$. The persona vector $v_{S,l}$ is defined as the difference between these vectors:

$$v_{S,l} = \mu_{S,l} - \mu_{Neutral,l}$$

This difference captures the latent direction associated with stylistic behaviour S, which can then be applied to steer model activations at inference time.

2.3 Using Persona Vectors to Effect Downstream Task Performance

Having extracted the layer-specific persona vectors, we measure their influence through activation steering on the Evaluation split.

At each decoding step t of the forward pass, we modify the residual stream activation $h_{t,l}$ at the target layer l by injecting the corresponding persona vector scaled by a steering coefficient α:

$$h'_{t,l} = h_{t,l} + \alpha \cdot v_{S,l}$$

Here α, controls the steering intensity, allowing us to study how varying levels of steering affects task performance.

3 Experimental setup

3.1 Datasets

We evaluate model performance across a set of multiple-choice benchmarks spanning domains such as mathematics, history, science, trivia common knowledge and reading comprehension.

- **MMLU Maths** We use the elementary mathematics subset of the Massive Multitask Language Understanding benchmark ($N = 500$ samples). (Hendrycks et al., 2021)
- **TruthfulQA** This dataset contains 817 questions designed to test common human misconceptions and trivia knowledge. (Lin et al., 2022)
- **RACE:** We use a subset of *RACE-middle* ($N = 500$ samples). These reading comprehension tasks extend beyond fact retrieval, requiring contextual understanding and reasoning over relationships. (Lai et al., 2017)
- **Mind Your Tone** Although smaller in size ($N = 50$), we utilize the Mind Your Tone dataset because it is the foundational work on the impact of query style on MCQ performance. This benchmark covers questions on mathematics, science, and history. (Dobariya and Kumar, 2025)

3.2 Models

We conduct our experiments using three instruction-tuned LLMs. These models were selected to balance reasoning capability with our computational constraints.

Phi-3-mini-4k-instruct (3.8B) (Abdin et al., 2024); a 32-layer,

Qwen3-4B-Instruct-2507 (4B) (Team, 2025); a 36-layer model,

OLMo-2-1B-Instruct (1B) (Groeneveld et al., 2024); 16-layer fully open-data model by the Allen Institute for AI. Although smaller in size, OLMo was specifically included as a fully open-data architecture.

3.3 Experiment design

Data Split Each dataset is split into a Steering set (20% for Mind Your Tone due to small size, 40% for rest of datasets) and the rest for Evaluation set. The Steering Set is utilized exclusively for persona vector extraction, where we capture internal activations as described in Section 2. The Evaluation Set is held out for inference-time steering, during which the previously extracted persona vector is used to steer the target layer and measure the effect on MCQ performance.

Layer l and α Sweeps We perform a grid search over both the transformer layers l and the steering intensity (α). Preliminary experiments revealed that steering early layers disrupted core linguistic capabilities, while later layers exhibited limited steerability. Accordingly, we focus our analysis on the lower-middle layers. Our goal is to identify the optimal layer-intensity pair (l, α) that maximizes MCQ accuracy for each persona.

3.4 Evaluation Metrics

We define the following quantitative metrics to assess the impact of steering on task performance:

3.4.1 Accuracy

The primary metric for multiple-choice question is accuracy. This is measured as the percentage of questions in the Evaluation split where the model's response (A, B, C, or D) matches the ground truth option.

This directly captures how the induced persona affects downstream task performance across our benchmarks.

3.4.2 Steering Influence on Entropy

To quantify how steering affects model's uncertainty, we calculate the entropy of the probability distribution over the top $k = 4$ tokens at the first generation step. A decrease in entropy indicates that the steering vector acts as a focusing mechanism, concentrating probability mass and increasing decisiveness. Conversely, an increase in entropy suggests that the persona introduces higher uncertainty.

3.4.3 Expected Calibration Error

To assess whether the model's confidence remains reliable under steering, we compute the Expected Calibration Error (Pavlovic, 2025). A model is considered calibrated if the predicted probability of an answer matches its frequency of correctness.

For N predictions partitioned into M bins by confidence, let Bin_m denote the set of predictions in bin. The ECE is calculated as the weighted average of the absolute difference between the bin accuracy ($acc(B_m)$) and the bin confidence ($conf(B_m)$):

$$ECE = \sum_{m=1}^{M} \frac{|Bin_m|}{N} |acc(Bin_m) - conf(Bin_m)|$$

In our multiple-choice setting, we calculate confidence by extracting the logits z for the tokens corresponding to multiple-choice options at the first generation step and compute softmax probabilities over the options. The confidence for a given prediction i is defined as:

$$P(y = i|x) = \frac{\exp(z_i)}{\sum_{j \in \{A,B,C,D\}} \exp(z_j)}$$

where x represents the stylized query. The bin confidence $\text{conf}(\text{Bin}_m)$ is then computed as the mean of these individual probabilities for all samples within the bin.

4 Results

4.1 Optimal Layers and Steering Intensity

In this section, we report and analyse the impact of activation steering on task performance. We begin by visualizing steering effectiveness as a function of layer l and steering coefficient α, focusing initially on the Mind Your Tone dataset on 3 best performing persona styles (Admiration, Insulting, Polite).

4.1.1 Model Phi-3

As illustrated in Figure 1, steering at layers 9–10 results in a slight performance degradation, whereas layers 12–13 are optimal, yielding accuracy gains of up to 12.5% over the baseline. Performance in later layers remains largely unchanged. This suggests that early layers are brittle at the tested steering strengths, while the final layers are sufficiently robust to remain unaffected. We conclude that effective steering is concentrated in the early-mid-layers layers, where the model is sufficiently steerable yet robust.

Regarding the strength in steering strength, we identified an optimal range of $\alpha \approx 10.0$–12.5 within the optimal layers (12–13). Although the exact optimum varies for each personality, we present results for three selected highest improvement styles (admiration, insulting, polite) in Figure 1.

While the results show a degree of brittleness, they demonstrate the effectiveness of our solution, purely identifying a target layer and applying steering can significantly improve test-time MCQ performance by up to 12.5%.

4.1.2 Entropy and Expected Calibration Error

To assess the model's internal uncertainty and prediction reliability we calculated the entropy (H_{top4}) and the Expected Calibration Error. This analysis focuses on Layer 12 of the Phi-3 model using the Mind Your Tone dataset, where steering effects were most pronounced.

Figure 2 illustrates the entropy dynamics at Layer 12. As α approaches the 10–12.5 range, entropy reaches its minimum, indicating that the steering vector acts as a focusing mechanism that constrains probability mass. Crucially, this peak in confidence coincides with the optimal multi-choice question performance. This suggests that the induced persona successfully sharpens the decision boundary around the correct answer.

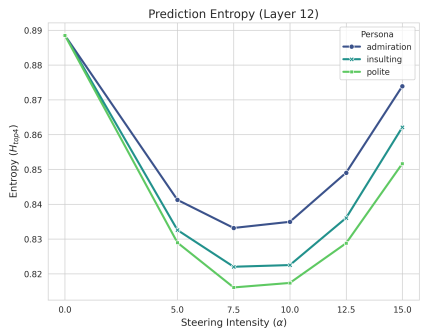

Figure 2: Entropy dynamics at Layer 12. The convergence of entropy to its minimum at the optimal α steering strength suggests that the steering vector acts as a focusing mechanism, maximizing model confidence at the point of peak performance.

Figure 1: Steering results for Model Phi-3 across three selected personalities (admiration, insulting, polite). Layers 12–13 show increased performance up to 12.5%, at an optimal steering strength of $\alpha = 10.0$–12.5. Values thresholded for at least 2.5% change

The Expected Calibration Error shown in Figure 3 follows a similar pattern, achieving minimal values at the optimal $\alpha \in [10, 12.5]$ range.

The importance of these results is that the simultaneous drop in both metrics indicates that the model is becoming both more confident and better calibrated.

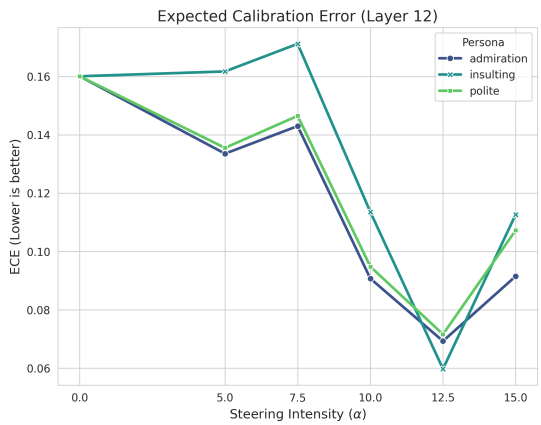

Figure 3: Expected Calibration Error at Layer 12. Minimal ECE aligns with peak model performance, suggesting model's confidence is well-calibrated.

4.1.3 Model Qwen

As shown in Figure 4, Qwen model achieves best performance when steered at layer 13. Interestingly, we observe a peaked distribution for the steering strength α: low values provide insufficient signal to alter the model's performance, while excessively high values degrade downstream model performance back to baseline.

4.1.4 Model OLMo

Steering results for OLMo model are shown in Figure 5. Unlike Phi-3 and Qwen models, OLMo displays inconsistent responses across adjacent layers and across persona styles. While layer 8 and 9 improves downstream performance for admiration style, it does not hold for insulting or polite styles.

A plausible explanation for this behaviour is the model's significantly smaller size (1B parameters) compared to Phi-3 and Qwen. We argue that the reduced parameter count likely results in a less disentangled latent space, making it difficult to isolate effective steering vectors.

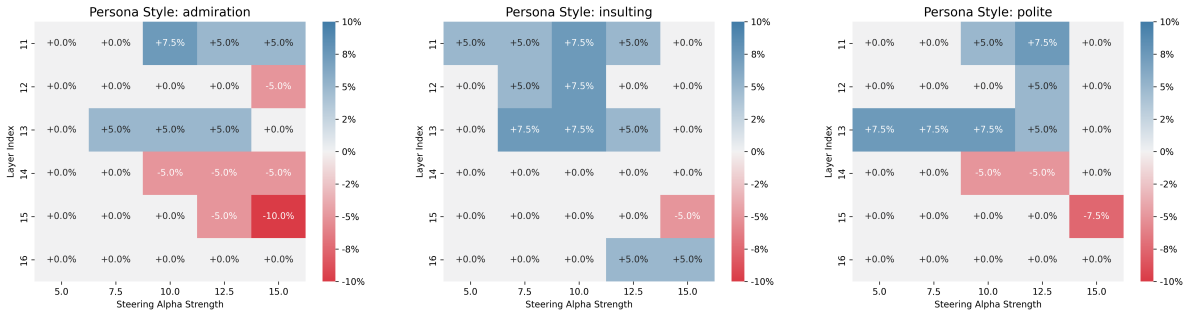

Figure 4: Steering dynamics for Model Qwen across three selected personalities (admiration, insulting, polite). Best accuracy achieved at Layer 13 with steering strength α characterized by a peaked pattern: accuracy gains are maximized at intermediate values, while extreme values (too low or too high) diminish the steering effect. Values are thresholded for at least 2.5% change.

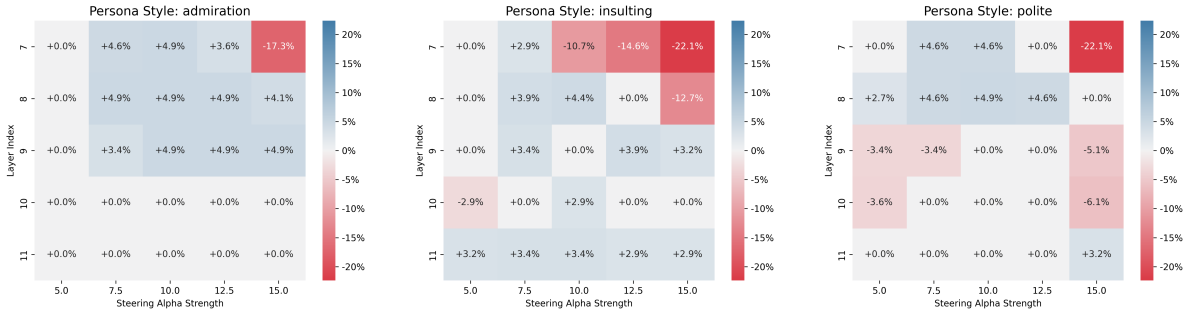

Figure 5: Steering dynamics for Model OLMo across three selected personalities. Values thresholded for at least 2.5% change.

4.2 Performance on MMLU, TruthfulQA, and RACE Datasets

Figure 6 illustrate the accuracy improvement for each model on the rest of the datasets, allowing us to identify the optimal steering layers. Previous results from the Mind Your Tone dataset suggested that lower-middle layers are the most effective targets for steering; our analysis of the MMLU, TruthfulQA, and RACE datasets confirms this observation.

Phi-3 demonstrates best accuracy improvement of 7.9% on MMLU Math (Layer 12) and 8.3% on TruthfulQA (Layer 11).

Similarly, Qwen achieves substantial 11.5% improvement on MMLU Math at Layer 15. However, its gains on TruthfulQA were only (1.7% at Layer 14).

In contrast, OLMo displayed a different profile. It showed negligible gains on MMLU Math, but yielded improvements on TruthfulQA (4.9% at Layer 7) and a substantial 10.7% increase on the RACE benchmark (Layer 10).

Notably, the effectiveness of steering on the RACE dataset appears highly dependent on baseline performance. Phi-3 and Qwen, which started with high baselines of 85.0% and 89.7% respectively, showed minimal improvement. Conversely, OLMo, which had a much lower baseline of 35.0%, benefited significantly from steering intervention.

4.3 Analysis of Optimal Persona Style

Previous sections focused on the parameters (layer l and α) for best steering effect on performance, in this section we analyse the results to see which persona styles were best for each model-dataset pair. Table 1 summarizes these findings showing optimal steering configurations: persona, layer, model and dataset configuration. In instances where multiple personas achieved same peak performance, we report the style that occurred across other benchmarks to prioritize generalization of persona styles.

For Phi-3, the *Nervous* persona emerged as the dominant configuration, best across two datasets (MMLU Math, and TruthfulQA). Qwen demonstrated a preference for various personas *Admira-*

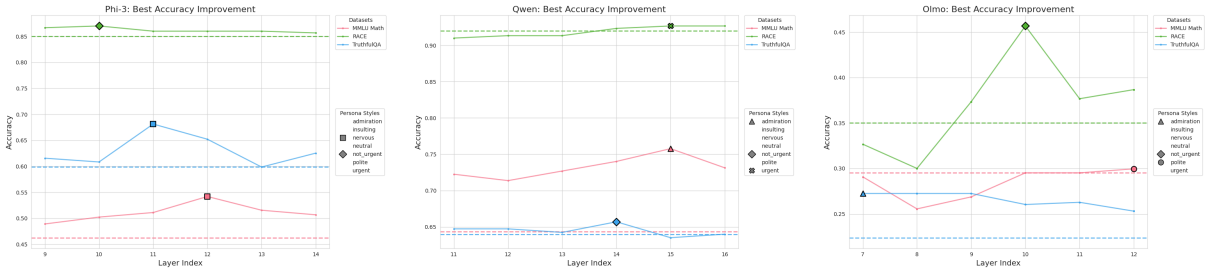

Figure 6: **Layer-wise steering performance analysis.** The figures illustrate the maximum accuracy improvement among persona styles (solid lines) versus baseline (not steered) performance (dashed lines) across MMLU, TruthfulQA, and RACE. The markers indicate the specific persona responsible for the best improvement. **(a)** Phi-3 shows improvement in layers 11–13. **(b)** Qwen peaks in layers 14–16, particularly on MMLU. **(c)** OLMo shows improvement only on RACE dataset (Layer 10), likely benefiting from the lower baseline.

tion, Polite, Urgent, Not Urgent. OLMo preferred *Admiration* for Mind Your Tone and TruthfulQA.

Our solution does not identify a single, "universal" persona that generalizes across all model-dataset pairs. Instead, our main contribution lies in empirically identifying the optimal steering parameter configuration for the introduced models and demonstrating that low-level stylistic attributes are significant for modulating downstream task performance.

Table 1: Summary of optimal persona style on performance. Results highlight model-dataset specific persona preferences, such as the dominance of the *Nervous* persona for Phi-3 or *Admiration* for OLMo. The lack of a universal persona across all models highlight the limitations of our method.

Model	Dataset	Persona	Layer	Gain
Phi-3	Mind Your Tone	Admiration	12	+12.5%
	MMLU Math	Nervous	11	+7.9%
	TruthfulQA	Nervous	12	+8.3%
	RACE	Not Urgent	10	+2.0%
Qwen	Mind Your Tone	Polite	13	+7.5%
	MMLU Math	Admiration	14	+11.5%
	TruthfulQA	Not Urgent	15	+1.7%
	RACE	Urgent	15	+0.1%
OLMo	Mind Your Tone	Admiration	8	+4.4%
	MMLU Math	Polite	7	+0.4%
	TruthfulQA	Admiration	12	+4.9%
	RACE	Not Urgent	10	+10.7%

4.4 Generalization

As described in Section 3.3, persona vectors are derived from a Steering split of a given benchmark dataset. To investigate the generalization of our method, we perform a "cross-steering" evaluation: we extract the persona vector using the Steering split of a "source" dataset and apply it to the evaluation split of a different "target" dataset. We conducted this experiment using the Phi-3 model across three datasets of comparable scale: MMLU Math, TruthfulQA, and RACE. For each dataset pair, we utilized the optimal layer l, α and personality style identified in our previous Section 4.2. Namely, MMLU Math (layer 12, Nervous), TruthfulQA (layer 11, Nervous), and RACE (layer 10, Not Urgent).

Our results indicate that datasets MMLU Math and TruthfulQA, demonstrate high mutual transferability. Steering TruthfulQA with a vector derived from MMLU Math achieved a performance gain of 6.15%, which is only slightly lower than the 8.28% gain achieved with "in-distribution" steering. The TruthfulQA-derived vector was effective on MMLU Math, achieving a 6.82% gain, nearly identical to the MMLU vector's performance. This suggests that the representation of a persona is not tied to the specifics of the questions, but represents a high-level shift that consistently influences the model's reasoning across different datasets.

Applying these vectors derived from MMLU Math or TruthfulQA to RACE showed limited transferability. Notably, steering vectors derived from RACE failed to hindered performance on the other datasets.

5 Limitations and Future Work

While our solution successfully increases the model's downstream task performance, its main

Table 2: Cross-dataset generalization of our steering method on Phi-3 model. Values denote **Accuracy (steered)** and (*Gain over Baseline (no steering)*). Grey cells indicate within-dataset steering; Green indicates successful transferability with improved performance compared to the no-steering base-line; Red indicates degraded transferability compared to the no-steering baseline.

	Target dataset (Evaluation)		
	MMLU Math	**TruthfulQA**	**RACE**
Source dataset (Vector Extraction)			
MMLU Math	**54.19%** (+7.94)	**66.00%** (+6.15)	**86.60%** (+1.60)
TruthfulQA	**54.18%** (+6.82)	**68.13%** (+8.28)	**84.66%** (-0.34)
RACE	**43.17%** (-3.08)	**55.47%** (-4.38)	**87.00%** (+2.00)

limitations are connected to the static definition of personality styles and the reliance on a manual search for optimal steering parameters.

Static definition of personality styles Currently, our solution is restricted to six pre-defined persona styles. While we mitigated the risk of prompt-specific linguistic cues by generating three distinct variations for each persona style prefix, even these remain fixed and do not entirely capture the full diversity of chat-based interaction.

Grid search Secondly, our implementation requires a manual grid search to identify the optimal layer l and steering strength α. Our results show that these optimal (l, α) combinations are model-dependent, which make it necessary for the optimal settings to be found individually for each model. We restricted our scope to single-layer interventions, future work should consider multi-layer steering as a combination of layers could yield even more robust results.

Dynamic Activation Steering As a possible extension of our work, future research could explore dynamic activation steering. This approach could involve computing the similarity between the internal activations elicited by an incoming query and a set of pre-computed steering vectors. By selecting the appropriate vector and steering strength, the solution could dynamically steer the model's response toward or away from specific persona styles with the aim of increasing task performance, moving from manual parameter selection to a more adaptive solution.

Mind Your Tone dataset size Lastly, the Mind Your Tone dataset is relatively small in size (N = 50). Nevertheless, we included it in our experiments as it remains a crucial dataset for establishing the impact of persona styles on model performance.

6 Conclusion

In this work, we have demonstrated an effective inference-time approach for improving Multiple-Choice Question accuracy through activation steering. By defining personality styles that control social, temporal, and emotional aspects of prompt styles, we have shown that a model's latent states can be modulated to enhance MCQ task performance. On Mind Your Tone, steering Phi-3 at lower-middle layers improves accuracy by up to 12.5% over the baseline. On larger benchmarks the best settings achieve gains of 7.9% on MMLU Math and 8.3% on TruthfulQA for Phi-3, and 11.5% on MMLU Math for Qwen.

Beyond accuracy, analysis of internal dynamics reveals that optimal steering reduces Entropy and Expected Calibration Error. This indicates that our solution does not only improve the model's performance, but acts as a focusing mechanism that decreases internal uncertainty and improves calibration. We showed that the extracted persona vectors are transferable between different knowledge domains, suggesting that these styles represent high-level behavioural shifts rather than task-specific shift.

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A Persona Style Prefixes

As described in Subsections 2.1, 2.2 to extract persona-specific vectors, we utilized three distinct prefix variations for each of the six stylistic categories. These prefixes were prepended to the original queries during the vector extraction phase.

Polite

- "Please help me with answering this question:"
- "Please be so kind as to answer this question:"
- "May I politely ask you to answer this question:"

Insulting

- "You are so intellectually slow and ***, just answer correctly for once:"
- "Do not screw this up, you ***, and answer this question:"
- "Try not to fail again, you ***, and answer this question:"

Admiration

- "You are incredibly smart, and I know you can answer this question:"
- "Your intelligence is astounding, so I am sure you can answer this question:"
- "You are a genius, and I know you can figure out the answer to this question:"

Urgent

- "Time is running out, answer this question immediately:"
- "Answer this question as fast as you can, no time to waste:"
- "Do not waste a single second, answer immediately this question:"

Not Urgent

- "Feel free to answer whenever it suits you:"
- "There is absolutely no rush, whenever you have a moment, answer:"
- "No pressure at all, take your time to answer this question:"

Nervous

- "I am struggling so much solving this, answer this question:"
- "I desperately need your help, answer this question:"
- "My life depends on this, I cannot solve it, answer this question:"